

MISSION 2 – GEOFENCING

2.1 Objective

To analyze drone behavior under different geofence conditions by implementing, testing, and toggling geofence activation, and interpreting the drone's response to boundary constraints.

2.2 Fence Parameters Used

- FENCE_ENABLE = 1 (activated) / 0 (deactivated)
- FENCE_TYPE = Polygon
- FENCE_ACTION = RTL
- Boundary: Polygon drawn around Tech Green

2.3 Four Required Scenarios

Case 1 – No Fence (Baseline)

Prior to enabling the geofence, FENCE_ENABLE was set to **0** in the parameter list. The drone was flown freely beyond the Tech Green boundary with no restrictions. No breach message appeared and no failsafe was triggered, confirming that geofencing was fully inactive. This establishes the baseline free-movement behavior for comparison.

Case 2 – Fence Activated

FENCE_ENABLE was set to **1** and the polygon boundary was loaded around Tech Green. The drone remained within the defined boundary during normal flight. No breach was triggered, confirming the fence was active and correctly constraining movement.

Case 3 – Waypoint Outside Fence (Breach Triggered)

A waypoint was commanded outside the polygon boundary while FENCE_ENABLE = **1**. Upon approaching the boundary, the drone triggered a geofence breach. The on-screen telemetry displayed a **breach warning message**, and the drone automatically initiated **RTL (Return to Launch)** without completing the out-of-bounds waypoint. This confirmed that FENCE_ACTION = RTL was correctly configured and executed.

Case 4 – Fence Deactivated (Freedom Restored)

FENCE_ENABLE was toggled back to **0** and the mode was changed to **AUTO** via the parameter panel. Following this change, the drone was able to navigate freely beyond the previous Tech Green boundary with no breach warnings and no RTL trigger. This confirms that setting FENCE_ENABLE = 0 and changing mode back to **AUTO** fully restores unrestricted movement.

2.4 Observations

- With FENCE_ENABLE = 0, the drone moved freely beyond Tech Green with no warnings — establishing a clear no-fence baseline.
- When the fence was active, the drone halted forward progress at the polygon boundary.
- Crossing the boundary while FENCE_ENABLE = 1 produced a visible **on-screen breach message** in the telemetry feed and immediately triggered the RTL failsafe.
- Toggling FENCE_ENABLE back to **0** restored full freedom of movement, with no breach warnings on subsequent flights outside the boundary.
- System responded predictably and consistently across all four cases.

2.5 Conclusion

Geofencing effectively restricts drone movement and enforces operational boundaries. The four test cases demonstrated a clear, repeatable relationship between the FENCE_ENABLE parameter and drone behavior: with the fence off, movement was unrestricted; with the fence on, any boundary violation triggered an immediate breach warning and RTL response. Toggling FENCE_ENABLE = 0 reliably restored full movement, confirming correct parameter-driven geofence control.